

DRESDEN, Germany

WMO: 104880

Lat: 51.1281N

Lon: 13.7544E

Elev: 232

StdP: 98.56

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.2	-8.2	-14.8	1.1	-10.0	-12.0	1.4	-6.9	11.2	4.2	10.3	3.3	3.2	70	0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	31.0	19.3	28.9	18.5	26.9	18.2	20.5	27.4	19.7	26.3	19.0	24.9	3.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.2	13.5	21.9	17.8	13.2	21.6	17.0	12.4	20.8	60.0	27.8	57.2	26.4	54.7	24.7	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	8.2	7.3	-13.8	34.2	4.1	2.2	-16.7	35.8	-19.1	37.1	-21.4	38.4	-24.4	40.0		
(5)			-14.1	22.6	4.0	1.1	-17.0	23.3	-19.3	24.0	-21.6	24.6	-24.5	25.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	0.2	0.8	4.8	10.3	13.8	17.6	19.8	18.9	15.2	10.4	5.7	2.4
	DBStd	7.95	4.96	4.66	4.61	4.22	4.07	3.72	3.51	3.36	3.44	3.93	3.83	4.34
	HDD10.0	1204	303	257	167	47	11	1	0	0	2	44	137	236
	HDD18.3	3202	561	490	420	242	149	56	24	32	106	248	380	493
	CDD10.0	1223	1	0	6	56	129	229	305	277	158	56	7	1
	CDD18.3	180	0	0	0	1	9	34	72	51	12	1	0	0
	CDH23.3	1533	0	0	0	10	84	318	627	418	73	2	0	0
	CDH26.7	461	0	0	0	1	14	87	213	137	11	0	0	0
Wind	WSAvg	3.6	4.3	4.0	4.0	3.5	3.4	3.1	3.0	2.9	3.2	3.6	4.0	4.3
Precipitation	PrecAvg	650	44	33	42	42	58	69	85	80	54	47	55	51
	PrecMax	886	122	71	119	125	152	203	189	233	149	160	119	166
	PrecMin	422	1	4	16	1	10	22	7	10	12	13	0	19
	PrecStd	118	24	17	25	26	34	36	45	51	28	31	28	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	12.1	19.0	24.9	28.2	31.9	34.0	33.8	28.2	22.9	16.0	12.8
		MCWB	9.6	7.8	10.9	14.2	17.6	20.0	19.0	19.4	17.9	15.4	12.9	9.4
	2%	DB	10.8	10.0	15.8	22.0	25.9	29.0	31.2	30.2	25.2	19.9	14.1	10.9
		MCWB	8.4	6.7	9.7	13.3	16.8	18.4	19.0	20.0	16.8	14.6	11.2	8.2
	5%	DB	8.8	8.2	13.2	20.0	23.2	27.0	29.0	27.2	23.2	18.0	12.2	9.2
		MCWB	7.0	5.6	8.6	12.7	15.2	18.0	18.4	18.6	16.4	13.5	9.7	7.3
	10%	DB	7.0	6.9	11.1	18.0	21.1	24.9	26.9	25.2	21.1	16.1	11.0	8.0
		MCWB	5.3	4.7	7.6	11.5	14.2	17.2	18.2	17.7	15.6	12.9	8.7	6.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	8.9	11.9	15.4	19.0	22.0	21.4	21.5	19.7	16.4	13.2	10.1
		MCDB	12.1	10.7	16.6	22.6	26.4	27.4	28.5	28.4	25.0	20.8	15.5	11.9
	2%	WB	8.2	7.2	10.4	14.1	17.6	20.1	20.2	20.3	18.3	15.3	11.4	8.6
		MCDB	10.3	9.2	14.5	20.3	23.9	27.0	27.3	27.1	23.3	18.8	13.8	10.5
	5%	WB	6.7	5.7	9.4	13.0	16.3	18.8	19.6	19.6	17.2	14.1	10.2	7.4
		MCDB	8.1	7.8	12.7	18.5	21.5	24.9	26.4	25.9	21.4	17.1	12.0	8.9
	10%	WB	5.2	4.5	7.9	12.1	15.3	17.7	18.9	18.7	16.1	13.0	9.1	6.3
		MCDB	6.5	6.5	10.7	17.4	20.0	23.2	25.0	23.8	20.1	15.8	10.8	7.7
Mean Daily Temperature Range	5% DB	MDBR	4.4	5.3	7.0	9.1	9.1	9.4	9.6	9.3	8.2	6.7	4.9	4.4
		MCDBR	6.1	7.6	11.0	12.4	12.4	12.9	13.2	12.8	11.2	9.4	6.5	5.8
		MCWBR	4.7	5.1	6.4	6.1	5.7	5.5	4.3	4.7	5.0	5.1	4.4	4.4
	5% WB	MCDBR	5.8	7.0	10.1	11.4	10.9	11.4	10.9	11.1	9.8	8.7	6.1	5.4
		MCWBR	4.7	5.1	6.2	6.0	5.5	5.4	4.4	4.8	5.0	5.2	4.5	4.3
Clear-Sky Solar Irradiance	taub	0.309	0.333	0.376	0.418	0.420	0.415	0.432	0.426	0.387	0.362	0.334	0.304	
	taud	2.362	2.338	2.250	2.183	2.218	2.260	2.221	2.255	2.338	2.385	2.402	2.381	
	Ebn at Noon	707	788	813	818	836	842	819	803	795	744	670	654	
	Edh at Noon	63	84	110	132	134	129	133	122	101	80	61	54	
All-Sky Solar Radiation	RadAvg	0.76	1.48	2.47	4.01	4.88	5.22	4.95	4.31	3.09	1.80	0.86	0.58	
	RadStd	0.12	0.24	0.33	0.53	0.58	0.43	0.59	0.40	0.41	0.23	0.12	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (52 neighbors)	+0.39	N/A	N/A	N/A	+0.35	+0.35	N/A	N/A	N/A	+18

Nomenclature: See separate page